9.10 Always Able to Give Change

WORKSHEET

The question

STEP I 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Each £1 customer gives you a coin; each £2 customer takes one away (as change). When do you get stuck?

STAGE 2 Two £2 coins

Ask yourself The first person must have a £1 coin. What can happen with the second person, and in each case, what situation are you left in?

Pause and try Try the same approach for $n = 3$. It works, but the cases multiply. Is there a way to count the successful queues all at once?

STAGE 3 Queues as paths

Ask yourself Draw the number of coins you hold against the number of people served. What does a queue look like? What does a successful queue look like?

STAGE 4 The reflection

Ask yourself Take a bad path. Find the first moment it touches -1, and reflect the part of the path *before* that moment in the line $y = -1$. Where does the new path start, and where does it end?

What's really going on

Catalan numbers

When $m = n$ (exactly enough £1 coins), the number of successful queues is

$$\binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n},$$

the nth *Catalan number* C_n: $1, 1, 2, 5, 14, 42, 132, \ldots$. Richard Stanley's book *Catalan Numbers* lists more than two hundred different things that they count. Among them: the number of ways to arrange n pairs of brackets correctly ("(()()" is allowed, ")()(()" is not, by exactly the coin argument with "(" as £1 and ")" as £2), the number of ways to cut a polygon with $n + 2$ sides into triangles by non-crossing diagonals, and the number of ways to bracket a product of $n + 1$ factors.

Random walks

Queues are random walks, and the reflection principle is one of the basic tools for studying them. It shows, for example, that a fair random walk starting at 0 is certain to return to 0 eventually, and that the probability that it stays strictly positive for the first $2n$ steps is $\frac{1}{2}\binom{2n}{n}2^{-2n} \approx \frac{1}{2\sqrt{\pi n}}$. In continuous form, the same reflection argument applied to *Brownian motion* gives the distribution of its maximum. This is the basis of the pricing of certain financial options, whose payoff depends on whether a share price ever crosses a barrier.

Bertrand's ballot problem

In 1887 Joseph Bertrand asked: in an election where candidate A receives a votes and B receives $b < a$, what is the probability that A is strictly ahead throughout the count? The answer, $\frac{a-b}{a+b}$, was proved in the same year by Désiré André, whose argument is usually remembered as the reflection principle (although historians have pointed out that André's actual proof used a slightly different bijection). Our raffle problem is a variant in which ties are allowed: success means the £1 count never falls *below* the £2 count. One of the exercises deduces Bertrand's answer from ours.

Try this next

1. **Three by cases.** Prove part (iii) by conditioning on the first people in the queue, as in part (ii), to see how much work the reflection principle saves.

2. **Bertrand's ballot.** In an election, A gets a votes and B gets $b < a$. Show that the probability that A is strictly ahead throughout the count is $\frac{a-b}{a+b}$. (*Hint:* the first vote must be for A; after that, A must never fall behind B by more than…)

3. **Brackets.** List the 5 correct arrangements of three pairs of brackets, and the 5 ways to cut a pentagon into triangles. Can you find a bijection between them?

4. **A £5 note.** A queue contains m people with £1 coins and n with £5 notes, for a ticket costing £1. Explain why the reflection principle no longer gives a simple answer. (This generalisation is much harder.)

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter III. The classic treatment of random walks and the reflection principle.
- **Richard Stanley, *Catalan Numbers*** (Cambridge University Press). Over two hundred interpretations, with exercises.
- **Marc Renault, "Lost (and found) in translation: André's actual method and its application to the generalized ballot problem"**, *American Mathematical Monthly* 115 (2008). The history of the reflection principle.

The full guided solution is in Section 9.10:
https://beyondstep.keithmcnulty.org/ch09/ballot.html